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Study on Limited Data Problem - Semantic Segmentation of Retinal Fundus Images

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Abstract

Medical images captured through various imaging modalities are key to the diagnosis and treatment of diseases, and these images serve as a major input to AI driven models but due to the scarcity of medical experts and their quality research time, the quantum of data required for AI models is limited. Moreover, medical image analysis is a laborious and error prone task. Early detection of retinal diseases by means of AI based semantic segmentation model has been boon to the diagnostic system. However, performing semantic segmentation with limited data of retinal fundus image is quite challenging. The present study is primarily focused on investigation of effectiveness of data augmentation with gamma corrected images to alleviate the problem of limited annotated data in deep learning models. The most widely accepted U-Net model in medical domain for image segmentation is trained employing retinal fundus image dataset augmented with gamma corrected images on all image channels and then tested against major publicly available datasets. The proposed method has outperformed other complex contemporary methods in terms of sensitivity and has also shown better generalizability across datasets from different institutions.

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1. Introduction

Medical imaging techniques are becoming important part of diagnosis and treatment in medical domain. Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Ultrasound Imaging (UI), and Positron-Emission Tomography (PET) are the widely used imaging technologies for detection and quantification of disease. However,

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the accurate diagnosis using medical images is a time consuming and tedious task and demands the involvement of medical experts. Deep learning models have shown an enormous potential in the field of computer for image classification, object detections, and segmentation. This has made them a popular choice for the segmentation of medical images for analysis and diagnosis of various diseases. Deep learning models are gaining popularity in the medical domain due to the scarcity of medical experts. These techniques also help to reduce human errors in medical diagnosis due to the tedious and repetitive nature of medical image analysis and can overcome individual experts' limitations[1]. The major reason behind the success of Deep Leaning models in Computer Vison (CV) is the availability of enormous, labelled data. However, models trained with limited data usually suffers from overfitting issues[2] and fails to generalize to the datasets from actual clinical settings. Limited annotated data is the major training problem in medical image analysis due to the privacy of patients and unavailability of experts for extensive annotation[3]. Like natural images in other datasets, medical images cannot be annotated through crowd sourcing and always needs a medical expert. Although limited annotated data is a prominent problem in medical imaging domain, it is also applicable to natural image domain because preparation of annotated data to relatively new area is always a difficult and time-consuming task.

Although, the limited annotated data problem has been addressed applying a lot of different approaches, an approach that is frequently utilized in deep learning-based models is data augmentation. The data augmentation approach generates a new set of images by applying various image transformation techniques such as geometric transformations, cropping, intensity operations etc.[4]. Primary objective of this proposed work is to investigate efficacy of data augmentation employing image enhancement techniques in limited annotated data constraint for segmentation of retinal fundus image segmentation using U-Net model[5].

2. Related Works

AI based Image analysis employing deep learning techniques in medical domain has two-fold issues, first is the limited availability of annotated data and next the quality of images differs across institutions due to different image acquisition settings. Besides providing solution for overfitting, data augmentation provides benefit in deep learning such as it provides accurate, inexpensive and controllable data for training[6].In Deep learning, application of basic data augmentation process is considered as fundamental step. However, in the literature very few evidence is available that explores the impact of image enhancement techniques on performance of the model. Olubunmi Sule[7] has made comprehensive survey of deep learning techniques for the segmentation of retinal fundus images. This article has concluded that the U-Net architecture is most promising architecture. Along with this the need of pre-processing of the images is also highlighted since the images from different datasets are captured with different camera setting and illumination level.

Paweł Liskowski et al.[8] has applied Global Contrast Normalization (GCN) and Zero-Phase Component Analysis (ZCA Whitening) followed by gamma correction on Saturation and value channel in HSV colour space for the pre-processing and augmentation of the retinal fundus images. This paper employs structured prediction formalism of conventional machine learning technique with convolutional neural network for feature extraction. Dasgupta et al.[9] also used similar technique with Contrast limited adaptive histogram equalization (CLAHE) technique followed by gamma correction. Xiuqin et al[10] used a RESUNET architecture that combines residual unit with U-NET model. This method also has adopted CLAHE method on green channel of the filtered and denoised images [11]. The complex network architecture and the image processing has resulted into a minor improvement in the accuracy of the model. Olubunmi Sule et al.[12] trained the U-Net model using gray scale retinal fundus images. The image quality has been improved by applying gamma correction on gray scale images followed by image enhancement using CLAHE.

Shorten et.al[13] assembled an exhaustive compilation of image augmentation methods for deep learning models. and highlighted the need of exploring the performance benchmarks across different data augmentation techniques under limited data problem. Xu Sun et al.[14] presented a retinal blood vessel segmentation applying channel wise gamma correction with morphological transformation. The result has shown the improvement in the performance, but it has impact of morphological transformation that modified the fine-grained blood vessels artificially. Also, the gamma correction in RGB colour space can alter the colour distribution of the medical images that may result into unwanted artifacts in the image. Histogram equalization and gamma corrections are the two important techniques used to enhance image quality. Histogram equalization may result into over-enhancement for frequent intensity levels. Rahman et al.[15] proposed an Automatic Gamma Correction method to enhance image contrast without creating any visual artifact and suggest procedure for selection of parameters of gamma correction

based on the statistical properties of image. The human vision system is more sensitive to intensity changes in the image[16], Consequently, the proposed method applies a gamma correction to the intensity channel within the HSI color space. RV_GAN[17] is essentially a novel multi-scale generative architecture that improves microvessel localization and segmentation by employing two generators and two multi-scale autoencoding discriminators. A weighted feature matching loss is introduced that incorporates and gives priority to features derived from the decoder of the discriminator rather than the encoder. Despite complex architecture, RV-GAN fails to improve the sensitivity of the model. In the literature, many proposed models[18],[19],[20],[21] has achieved better sensitivity but at the cost of other performance degradation. Vessel graph network(VGN) [20] has achieved sensitivity of 0.9382 but suffers from increased false positives. Many other methods such as gamma correction a preprocessing method[22] and Parallel programming approaches[23] have been explored to improve the performance of AI based solutions in medical image processing.

Despite the architectural development in deep learning models for image segmentation, the key challenges are these models still suffer from generalizability of the model across different quality of images and overfitting due to limited annotated training data. It is also an important requirement to study the implication of image quality enhancement on overall generalizability and performance under limited annotated data training.

Through systematic study of reported latest literatures and potential models, the following research gaps are identified. Firstly, there is need for experimenting in improving the quality of images by some means of pre-processing such as gamma correction or histogram equalization evaluating the impact on the performance of the model. This will eventually improve the performance of segmentation with limited data. Secondly, Study by employing dataset from different sources during training and testing will be of great importance.

3. Proposed Methodology

A primary aim of this study is to enhance the segmentation capabilities of the U-Net model, which was trained using limited annotated dataset, by augmenting a collection of gamma-corrected images. The architecture of the system is as shown in Figure 1.

Here, The U-Net[5] model has been used for the segmentation of images because it is the most widely used model in the medical domain. The U-Net is a supervised segmentation model and trained using input images along with associated segmented images for desired segmentation outcome. The model learns necessary image features for segmentation during training phase and produces segmented output image for the given input image during the testing phase. The complete methodology has been described in the following sections.

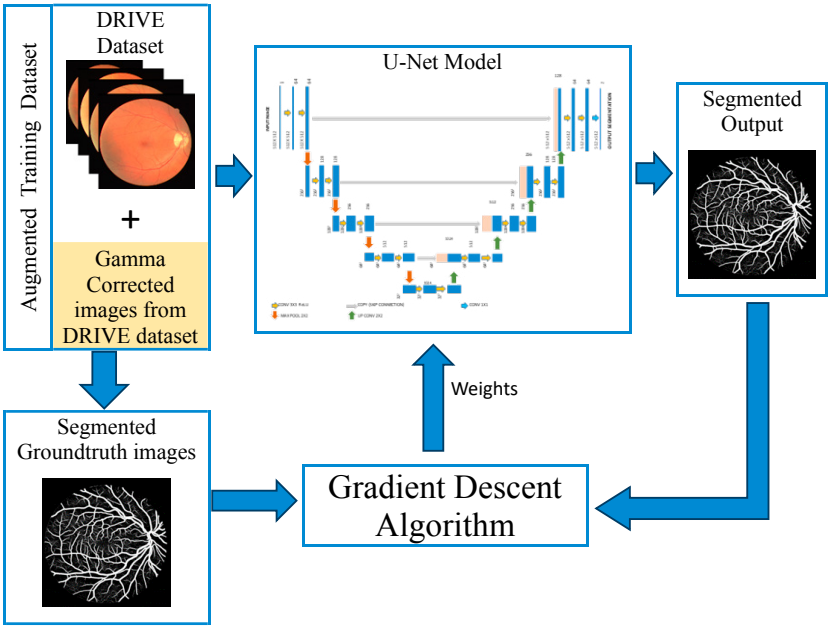


Figure. 1: U-Net Model training with augmented gamma corrected image dataset

3.1. Gamma (γ) Correction

The basic gamma correction equation is defined as:

$$I_{out} = cI_{in}^{\gamma} \tag{1}$$

Where, I_{out} is the gamma corrected output image, I_{in} is the input image, c and γ are the constants that depends on statistical properties of an image. The value of c is set to 1 and γ is to 0.5,1 and 1.25. In the proposed method, values of c and γ are pre-calculated since the training images are not going to be changed dynamically. Images are first converted to HSI colour space[24], then the intensity channel is corrected using equation no. 1 and converted back to RGB colour space. The effectiveness of colour space transformation is evaluated by performing gamma correction in both RGB and HSI colour space.

3.2. *Dataset:* The openly available medical Image datasets used for this work are as follows:

Table 1: Datasets available for segmentation of retinal fundus images.

Sr.No.	Dataset	Organization	Size (width X height	Number of images
1	Digital Retinal Images for Vessel Extraction (DRIVE)[25]	screening program for diabetic retinopathy, Netherlands	768 x 584	40
2	Structured Analysis of the Retina (STARE)[26]	University of California, San Diego	700 x 605	20
3	Child Heart and Health Study in England Dataset(CHASE_DB)[27]	Child Heart and Health Study in England (CHASE)	1280 x 960	28
4	High-Resolution Fundus (HRF) Image Dataset[28]	Brno University of Technology, Brno, Czech Republic and	3504 x 2336	30

		University of Erlangen, Nuremberg, Germany		

3.3. Training and Testing of the model:

The primary objective of this work is to investigate the performance of the U-Net model for the segmentation of medical images under limited annotated data constraints. The model is trained using the twenty real images from DRIVE dataset and further augmented by gamma corrected images with gamma values of 0.5,0.75, and 1.25 in HSI and RGB colour space. Five augmented datasets are created by mixing the twenty gamma corrected images with twenty real images from DRIVE dataset. No other augmentation technique is applied during training to realize the effectiveness of gamma correction method. The trained model is tested using images from DRIVE, STARE, and CHASE_DB datasets without applying any colour space transformation. Different training scenarios are compared using AUC_ROC and F1-score metric. Also, no post processing on the segmentation result is applied to improve the segmentation accuracy beyond the result of the model.

3.4. Performance Metrics:

The Image segmentation classifies every image pixel into distinct categories. In the binary segmentation, typical image pixels are classified as background and the desired object pixel. In the proposed work, the segmented retinal fundus image from the U-Net model is compared with the expert segmented corresponding blood vessel image. The background of the image is represented by class 0 and the blood vessels are represented by class 1. The performance parameters are calculated using confusion matrix[29] as shown in table 2, created using class labels of segmented image.

Table 2: Confusion Matrix

	Actual Pixel = 1 (Vessel Pixel)	Actual Pixel = 0 (Background Pixel)
Predicted as 1 (Vessel Pixel)	True Positive (TP)	False Positive (FP)
Predicted as 0 (Background Pixel)	False Negative (FN)	True Negative (TN)

The performance of the model is measured using precision, sensitivity, accuracy, and f1-score[31] and calculated using confusion matrix using equation (2).

$$Precision = \frac{TP}{TP + FP} \qquad Accuracy = \frac{TN + TP}{TN + TP + FP + FN} \tag{2}$$

$$Sensitivity\ or\ Recall = \frac{TP}{TP + FN} \qquad F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall}$$

4. Experiments and Results

Experiments are designed to find solutions to following research questions which are unanswered in current literatures.

Research Question 1: To investigate the influence of gamma corrected images during training on performance of segmentation of blood vessels in retinal fundus images.
Research Question 2: To investigate the generalizability of the model trained with dataset augmented with gamma corrected images.

The U-Net model is implemented using TensorFlow library in python. The model is trained using stochastic gradient descent optimiser having learning rate of 1E-04 with binary cross-entropy loss. It is trained up to 100 epochs. A server equipped with Intel Xeon 12 core processor, 64GB RAM and NVIDIA Tesla T4 graphics card is used for execution of the model.

The U-Net model is trained using dataset combinations shown in table no 2 to 5. Gamma correction is applied in two approaches. In first approach, original image is converted to HSI format and in only "Intensity" channel, gamma correction is applied. In second approach gamma correction is applied in all three 3 channels of RGB. The complete training and testing pattern is presented in Table.2. The table presents record of observation in terms of accuracy.

Through experiments, it is observed that there is no much significant difference among variants of Gamma corrected images. Hence record of observation will have only DRIVE20, CHASE, HRF and STARE as columns. This is reflected in Table 2 and thereafter. The training is performed using "adam" optimizer with learning rate of 1e-3, and binary cross entropy loss function. No additional image transformation has been applied to augment the dataset to focus on the impact of gamma correction-based augmentation. Also, even though other datasets are available for training, these are not used for training to emphasize on training with limited annotated dataset. However, the trained model is tested on all the openly available real images datasets. This experimentation also helps to analyse the generalizability of the model.

4.1 *Training and Testing performance*

Figure 2 presents performance during training and testing phase. Figures representing performance during training and testing with i) DRIVE20 dataset as well as testing with ii) DRIVE dataset with intensity level gamma correction is presented below.

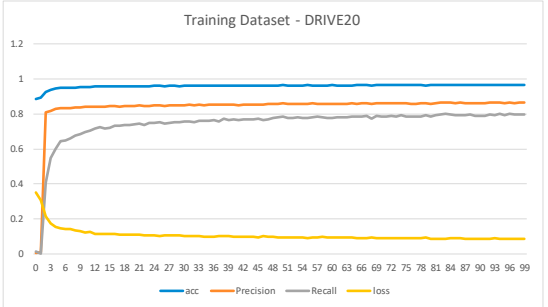


Figure 2.a: Performance during training with DRIVE 20 dataset

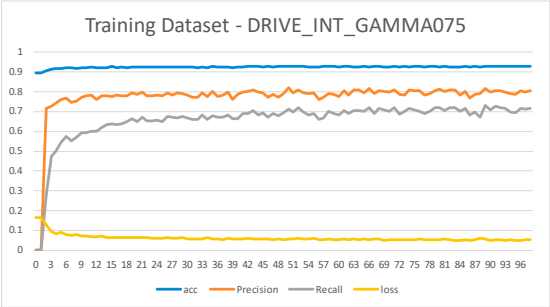


Figure 2.b: Performance during training with DRIVE INT with Gamma 0.75 corrected dataset

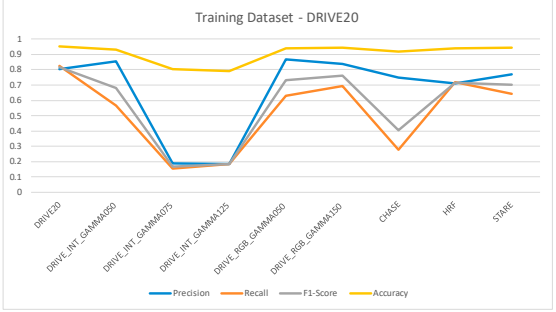


Figure 2.c: Performance during Testing with DRIVE 20 dataset

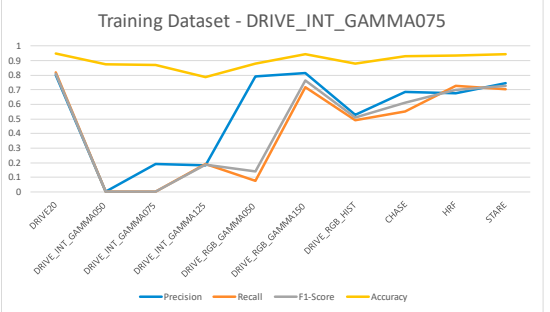


Figure 2.d: Performance during Testing with DRIVE INT with Gamma 0.75 corrected dataset

The performance of the model during testing, with respect to F1-score metric is shown in Table 3, performance with respect to accuracy is presented in Table 4, and sensitivity in Table 5.

Table 3: Model testing performance using **F1-Score.**					
Training Datasets	Testing Datasets				
	DRIVE20	CHASE	HRF	STARE	Coefficient of Variance
DRIVE20	0.8153	0.4060	0.7150	0.7006	0.268
DRIVE_INT_GAMMA0.50	0.8023	0.5381	0.7076	0.7075	0.160
DRIVE_INT_GAMMA0.75	**0.8101**	**0.6110**	**0.6995**	**0.7252**	**0.115**
DRIVE_INT_GAMMA1.25	0.6539	0.1105	0.5850	0.5043	0.525
DRIVE_RGB_GAMMA0.50	0.7956	0.4469	0.6946	0.6776	0.225
DRIVE_RGB_GAMMA0.75	0.7520	0.2576	0.6019	0.5986	0.897
DRIVE_RGB_GAMMA1.25	0.6939	0.0151	0.1052	0.3917	0.495

Table 4: Model testing performance using **Accuracy**					
Training Datasets	Testing Datasets				
	DRIVE20	CHASE	HRF	STARE	Coefficient of Variance
DRIVE20	0.9513	0.9183	0.9387	0.9434	0.015
DRIVE_INT_GAMMA050	0.9505	0.9262	0.9379	0.9448	0.011
DRIVE_INT_GAMMA075	**0.9511**	**0.9298**	**0.9336**	**0.9450**	**0.011**
DRIVE_INT_GAMMA125	0.9294	0.9044	0.9238	0.9251	0.012
DRIVE_RGB_GAMMA050	0.9514	0.9205	0.9380	0.9428	0.014
DRIVE_RGB_GAMMA150	0.9395	0.8997	0.9027	0.9170	0.020

Table 5: Model testing performance using **Sensitivity**					
Training Datasets	Testing Datasets				
	DRIVE20	CHASE	HRF	STARE	Coefficient of Variance
DRIVE20	0.8265	0.2786	0.7210	0.6429	0.386
DRIVE_INT_GAMMA050	0.7898	0.4290	0.7054	0.6479	0.240
DRIVE_INT_GAMMA075	**0.8187**	**0.5503**	**0.7254**	**0.7049**	**0.159**
DRIVE_INT_GAMMA125	0.5239	0.0593	0.5040	0.3699	0.589
DRIVE_RGB_GAMMA050	0.7440	0.3207	0.6614	0.5837	0.317
DRIVE_RGB_GAMMA150	0.6505	0.0005	0.1524	0.2403	1.065
DRIVE_RGB_HIST	0.7110	0.1000	0.3658	0.4697	0.615

From this statistical analysis it is observed that a UNet Model trained with DRIVE20 dataset, augmented with Gamma corrected (Gamma=0.75) performs better and consistent across all dataset. This model has recorded accuracy of 0.945±0.0009, F1 score of 0.711±0.082 and Sensitivity of 0.699±0.111. This proves two aspects; first one is that augmentation by means of gamma correction has significant influence on improving performance of segmentation of blood vessel in retinal fundus images. Second aspect is Gamma corrected (0.75) model is performing consistently with all other dataset, depicting generalizability.

The majority of acquired images are of poor quality due primarily to the quality limitations of the acquisition system; as a result, they fail to depict the essential characteristics of the objects. Gamma correction has been instrumental in addressing this issue. It's a method of employing nonlinear filter to adjust brightness of image based on the predefined Gamma parameter. $\gamma<1$, enhances the brightness, from table it appears that increasing the brightness has resulted in enhancing visibility of diagnostic structures, and hence improved performance.

The segmentation result for the DRIVE_INT_GAMMA0.75 training dataset is as shown in Figure 3.

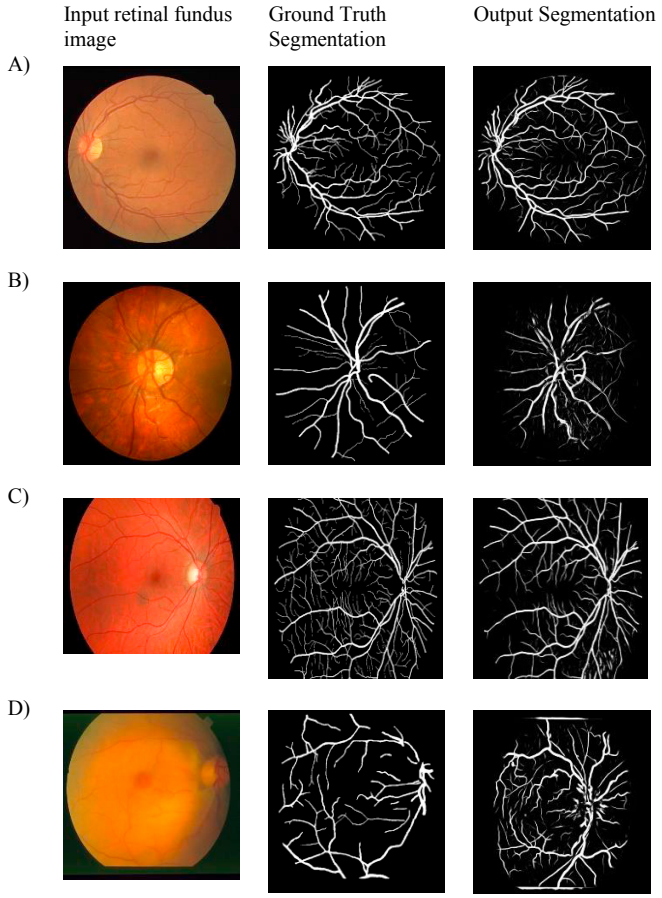


Figure 3: Segmentation Result
Dataset: A) DRIVE B)CHASE C)HRF D)STARE

4.2 *Qualitative Analysis*

In order to qualitatively evaluate the blood vessel segmented images, a comparison is made between the outcomes of the proposed gamma-corrected UNet-based model and those of other well-known supervised learning methods. Totally 5 medical experts were employed for this task. Each expert is presented with blood vessel segmented image from proposed model as well as from the Basu et al.[29], RV_GAN[17], and FR_UNet[30]. Provision to zoom out and zoom in the segmented image is provided. Experts were expected to rate in the Likert scale (1 to 5) with respect presence of anatomical structure of the retina such as an optic disc, retinal vasculatures, macula, and fovea in the segmented image. The diagnostic importance parameters like, presence, connectivity, rupturing of blood vessels are assessed with the help of medical expert. Following figure 4 presents the consolidated qualitative analysis made by medical expert.

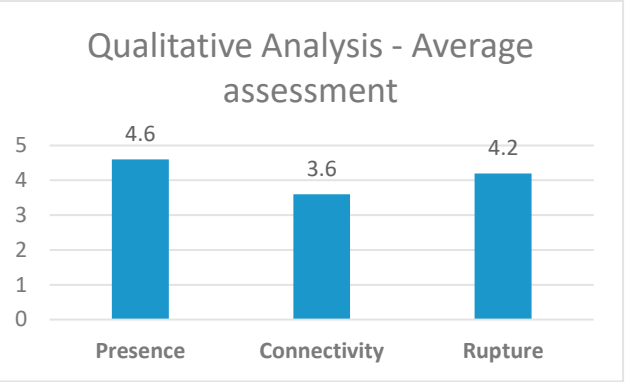


Figure 4: Qualitative analysis

Comparative analysis

The Table.5 presents the comparison of performance for segmentation without and with gamma corrections. The table also presents the comparison of proposed method with other contemporary methods

Table 6 : Comparative analysis

Sr.No.	Segmentation method	Accuracy	Sensitivity	AUC ROC	F1-Score
1	Segmentation with gamma (0.75) correction on Intensity	0.9511	0.8187	0.9737	0.8101
2	Segmentation with gamma (0.50) correction on RGB	0.9514	0.7440	0.9653	0.7956
3	Basu et al.[29]	0.9593	0.7119	--	0.75
4	RV-GAN[17]	0.9790	0.7927	0.9887	0.8690
5	Xu Sun et al. CWRGC[14]	0.9539	0.8140	0.9777	0.8174
6	Paweł Liskowski et al. with GCN[8]	0.9487	0.7550	0.9708	--
7	Paweł Liskowski et al. with ZCA[8]	0.9485	0.7819	0.9719	--
8	Dasgupta et al.[9]	0.9533	0.7691	0.9744	0.8074
9	RESUNET[10]	0.9650	0.8350	0.9811	--
10	Olubunmi Sule et al.[12]	0.9447	0.7092	0.9721	--
11	VSSC Net[19]	0.9627	0.7827	0.9789	--

From the Table 6, it is observed that Proposed method with gamma correction as preprocessing has performance scores which are on par with recently reported models. However, one unique observation that can be drawn here is sensitivity of proposed method (Gamma 0.75_INT) is better compared to other recently reported methods. Exclusively in case of segmentation sensitivity plays vital role. It indicates precision of vessel extraction.

5. Conclusion and Future Work:

The primary objective of this work is to analyse the performance of the U-Net model trained under the limited annotated dataset augmented with the gamma-corrected images. Results clearly indicate that the gamma-correction shows improvement in the performance for the gamma value of 0.75. The model performance drops at lower gamma due to the loss of contrast in the images and at gamma more than 1.0 due to the darker images. The primary reason for opting the gamma correction method for the correction of image quality because the medical images taken at laboratories differ in quality due distinct camera setup and illumination level. The segmentation performance for the CHASE dataset is lower than the other dataset due to its poor image quality. The variation in performance due to different gamma level also indicate that there will be unique gamma level for a dataset at which the performance will be at peak. However, the estimation of optimal gamma level for a dataset is very hard and even the adaptive gamma correction is also largely empirical. This experimentation also proved that the gamma correction on intensity channel of HSI colour space is more useful than the gamma correction on RGB channel. Future improvement to this work can

be identification of the optimal quality parameters for the images that can provide precise gamma correction to the image dataset and investigate the performance of model with different hyperparameters and loss functions.

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